EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. $\begin{vmatrix} y+z & x & x \\ y & z+x & y \\ z & z & x+y \end{vmatrix}$ equals -

(A) $x^2y^2z^2$ (B) $4x^2y^2z^2$ (C) xyz (D) $4xyz$

MDT124

2. If $\begin{vmatrix} 1 & 3 & 4 \\ 1 & x-1 & 2x+2 \\ 2 & 5 & 9 \end{vmatrix} = 0$, then x is equal to-

(A) 2 (B) 1 (C) 4 (D) 0

MDT125

3. If $px^4 + qx^3 + rx^2 + sx + t = \begin{vmatrix} x^2+3x & x-1 & x+3 \\ x+1 & 2-x & x-3 \\ x-3 & x+4 & 3x \end{vmatrix}$ then t is equal to -

(A) 33 (B) 0 (C) 21 (D) none

MDT126

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4. There are two numbers x making the value of the determinant $\begin{vmatrix} 1 & -2 & 5 \\ 2 & x & -1 \\ 0 & 4 & 2x \end{vmatrix}$ equal to 86. The sum of these two numbers, is-

(A) –4 (B) 5 (C) –3 (D) 9

MDT127

5. If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and A_2, B_2, C_2 are respectively cofactors of a_2, b_2, c_2 then $a_1A_2 + b_1B_2 + c_1C_2$ is equal to -

(A) $-\Delta$ (B) 0 (C) Δ (D) none of these

MDT128

6. If in the determinant $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, A_1, B_1, C_1 etc. be the co-factors of a_1, b_1, c_1 etc., then which of the following relations is incorrect-

(A) $a_1A_1 + b_1B_1 + c_1C_1 = \Delta$ (B) $a_2A_2 + b_2B_2 + c_2C_2 = \Delta$

(C) $a_3A_3 + b_3B_3 + c_3C_3 = \Delta$ (D) $a_1A_2 + b_1B_2 + c_1C_2 = \Delta$

MDT129

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7. If $\Delta=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and A_1, B_1, C_1 denote the co-factors of a_1, b_1, c_1 respectively, then the value of the determinant $\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$ is -

(A) Δ (B) Δ^2 (C) Δ^3 (D) 0

MDT130

8. If a, b, c are in AP, then $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix}$ equals -

(A) $a + b + c$ (B) $x + a + b + c$ (C) 0 (D) none of these

MDT131

9. For positive numbers x, y and z, the numerical value of the determinant $\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix}$ is -

(A) 0 (B) $\log xyz$ (C) $\log(x + y + z)$ (D) $\log x \log y \log z$

MDT132

10. Let a determinant is given by $A = \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$ and suppose $A = 6$. If $B = \begin{vmatrix} p+x & q+y & r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{vmatrix}$ then

(A) $B = 6$ (B) $B = -6$ (C) $B = 12$ (D) $B = -12$

MDT133

11. The value of an odd order determinant in which $a_{ij} + a_{ji} = 0 \ \forall \ i, j$ is -

(A) perfect square (B) negative (C) ± 1 (D) 0

MDT134

12. If $S_r = \begin{vmatrix} 2r & x & n(n+1) \\ 6r^2-1 & y & n^2(2n+3) \\ 4r^3-2nr & z & n^3(n+1) \end{vmatrix}$, then $\sum_{r=1}^{n} S_r$ does not depend on -

(A) x (B) y (C) n (D) all of these

MDT135

13. If $a, b, c > 0$ and $x, y, z \in R$, then the determinant $\begin{vmatrix} (a^x+a^{-x})^2 & (a^x-a^{-x})^2 & 1 \\ (b^y+b^{-y})^2 & (b^y-b^{-y})^2 & 1 \\ (c^z+c^{-z})^2 & (c^z-c^{-z})^2 & 1 \end{vmatrix}$ is equal to -

(A) $a^x b^y c^x$ (B) $a^{-x} b^{-y} c^{-z}$ (C) $a^{2x} b^{2y} c^{2z}$ (D) zero

MDT136

14. If a, b, c are sides of a scalene triangle, then the value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is :

(A) non-negative (B) negative (C) positive (D) non-positive

MDT137

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15. Let $f(x)=\begin{vmatrix}1+\sin^2 x & \cos^2 x & 4\sin 2x\\ \sin^2 x & 1+\cos^2 x & 4\sin 2x\\ \sin^2 x & \cos^2 x & 1+4\sin 2x\end{vmatrix}$, then the maximum value of $f(x)$, is-

(A) 2 (B) 4 (C) 6 (D) 8

MDT138

16. If the determinant $\begin{vmatrix}a+p & \ell+x & u+f\\ b+q & m+y & v+g\\ c+r & n+z & w+h\end{vmatrix}$ splits into exactly K determinants of order 3, each element of which contains only one term, then the value of K, is-

(A) 6 (B) 8 (C) 9 (D) 12

MDT139

17. Let $D_1 = \begin{vmatrix}a & b & a+b\\ c & d & c+d\\ a & b & a-b\end{vmatrix}$ and $D_2 = \begin{vmatrix}a & c & a+c\\ b & d & b+d\\ a & c & a+b+c\end{vmatrix}$ then the value of $\frac{D_1}{D_2}$ where $b \neq 0$ and $ad \neq bc$, is

(A) – 2 (B) 0 (C) – 2b (D) 2b

MDT140

18. If $a^2 + b^2 + c^2 = -2$ and $f(x)=\begin{vmatrix}1+a^2x & (1+b^2)x & (1+c^2)x\\ (1+a^2)x & 1+b^2x & (1+c^2)x\\ (1+a^2)x & (1+b^2)x & 1+c^2x\end{vmatrix}$ then $f(x)$ is a polynomial of degree-

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(A) 0 (B) 1 (C) 2 (D) 3

MDT141

19. The number of real values of x satisfying $\begin{vmatrix}x & 3x+2 & 2x-1\\ 2x-1 & 4x & 3x+1\\ 7x-2 & 17x+6 & 12x-1\end{vmatrix} = 0$ is -

(A) 3 (B) 0 (C) 1 (D) infinite

MDT142

20. The determinant $\begin{vmatrix}\cos(\theta+\phi) & -\sin(\theta+\phi) & \cos 2\phi\\ \sin\theta & \cos\theta & \sin\phi\\ -\cos\theta & \sin\theta & \cos\phi\end{vmatrix}$ is -

(A) 0
(B) independent of θ
(C) independent of ϕ
(D) independent of θ & ϕ both

MDT143

21. $\begin{vmatrix}(x^2+1)^2 & (xy+1)^2 & (xz+1)^2\\ (xy+1)^2 & (y^2+1)^2 & (yz+1)^2\\ (xz+1)^2 & (yz+1)^2 & (z^2+1)^2\end{vmatrix} = k(x-y)^2(y-z)^2(z-x)^2$, then k =

(A) 1 (B) 2 (C) 3 (D) 4

MDT144

For More Join: @IITJEE_Advanced

22. $D=\begin{vmatrix}10^4+2 & 10^7+3 & 10^8+8\\ 10^9+9 & 10^2+8 & 10^3-4\\ 10^3-5 & 10^8+b & 10^6+a\end{vmatrix}$ where a, b, both $\in \{1,2,3,4,5,6,7,8,9\}$

Number of ordered pairs (a, b) such that $D = 2n + 1, n \in Z$ is

(A) 36 (B) 20 (C) 16 (D) 45

MDT145

23. For $\alpha, \beta \in R$ and a natural number n, let $A_r=\begin{vmatrix} r & 1 & \frac{n^2}{2}+\alpha \\ 2r & 2 & n^2-\beta \\ 3r-2 & 3 & \frac{n(3n-1)}{2}\end{vmatrix}$. Then $2A_{10} - A_8$ is

(A) $4\alpha + 2\beta$ (B) $2\alpha + 4\beta$ (C) $2n$ (D) 0

MDT146

24. The value of k for which the set of equations $3x + ky - 2z = 0$, $x + ky + 3z = 0$ and $2x + 3y - 4z = 0$ has a non-trivial solution is-

(A) 15 (B) 16 (C) 31/2 (D) 33/2

MDT147

25. If the system of linear equations

$x_1 + 2x_2 + 3x_3 = 6$

$x_1 + 3x_2 + 5x_3 = 9$

$2x_1 + 5x_2 + ax_3 = b$

is consistent and has infinite number of solutions, then :-

(A) $a \in R - \{8\}$ and $b \in R - \{15\}$

(B) $a = 8$, b can be any real number

(C) $a = 8, b = 15$

(D) $b = 15$, a can be any real number

MDT148

26. Consider the system of equations : $x + ay = 0$, $y + az = 0$ and $z + ax = 0$. Then the set of all real values of 'a' for which the system has a unique solution is :

(A) $\{1, -1\}$ (B) $R - \{-1\}$ (C) $\{1, 0, -1\}$ (D) $R - \{1\}$

MDT149

27. Let a, b, c be any real numbers. Suppose that there are real numbers x, y, z not all zero such that $x = cy + bz$, $y = az + cx$ and $z = bx + ay$, then $a^2 + b^2 + c^2 + 2abc$ is equal to

(A) 2 (B) –1 (C) 0 (D) 1

MDT150

28. If the system of equation, $a^2x - ay = 1 - a$ & $bx + (3 - 2b)y = 3 + a$ possess a unique solution $x = 1$, $y = 1$ than :

(A) $a = 1; b = -1$ (B) $a = -1, b = 1$ (C) $a = 0, b = 0$ (D) none

MDT151

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For More Join: @IITJEE_Advanced

29. If the system of equation

$2x + y - z = 5$

$2x - 5y + \lambda z = \mu$

$x + 2y - 5z = 7$

has infinitely many solutions, then $(\lambda + \mu)^2 + (\lambda - \mu)^2$ is equal to

(A) 916 (B) 912 (C) 920 (D) 904

MDT152

30. The number of $\theta \in (0, 4\pi)$ for which the system of linear equations

$3(\sin 3\theta)x - y + z = 2$

$3(\cos 2\theta)x + 4y + 3z = 3$

$6x + 7y + 7z = 9$

has no solution is :

(A) 6 (B) 7 (C) 8 (D) 9

MDT153

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. The determinant $\begin{vmatrix} a^2 & a^2-(b-c)^2 & bc \\ b^2 & b^2-(c-a)^2 & ca \\ c^2 & c^2-(a-b)^2 & ab \end{vmatrix}$ is divisible by -

(A) $a + b + c$ (B) $(a + b)\,(b + c)\,(c + a)$

(C) $a^2 + b^2 + c^2$ (D) $(a - b)(b - c)\,(c - a)$

MDT154

2. The value of θ lying between $-\frac{\pi}{4}$ & $\frac{\pi}{2}$ and $0 \le A \le \frac{\pi}{2}$ and satisfying the equation

$\begin{vmatrix} 1+\sin^2 A & \cos^2 A & 2\sin 4\theta \\ \sin^2 A & 1+\cos^2 A & 2\sin 4\theta \\ \sin^2 A & \cos^2 A & 1+2\sin 4\theta \end{vmatrix} = 0$ are -

(A) $A=\frac{\pi}{4},\ \theta=-\frac{\pi}{8}$ (B) $A=\frac{3\pi}{8}=\theta$ (C) $A=\frac{\pi}{5},\ \theta=-\frac{\pi}{8}$ (D) $A=\frac{\pi}{6},\ \theta=\frac{3\pi}{8}$

MDT155

3. Which of the following determinant(s) vanish(es) ?

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(A) $\begin{vmatrix} 1 & bc & bc(b+c) \\ 1 & ca & ca(c+a) \\ 1 & ab & ab(a+b) \end{vmatrix}$ (B) $\begin{vmatrix} 1 & ab & \frac{1}{a}+\frac{1}{b} \\ 1 & bc & \frac{1}{b}+\frac{1}{c} \\ 1 & ca & \frac{1}{c}+\frac{1}{a} \end{vmatrix}$

(C) $\begin{vmatrix} 0 & a-b & a-c \\ b-a & 0 & b-c \\ c-a & c-b & 0 \end{vmatrix}$ (D) $\begin{vmatrix} \log_x xyz & \log_x y & \log_x z \\ \log_y xyz & 1 & \log_y z \\ \log_z xyz & \log_z y & 1 \end{vmatrix}$

MDT156

4. The determinant $\begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix}$ is equal to zero, if -

(A) a, b, c are in AP

(B) a, b, c are in GP

(C) α is a root of the equation $ax^2 + bx + c = 0$

(D) $(x - \alpha)$ is a factor of $ax^2 + 2bx + c$

MDT157

For More Join: @IITJEE_Advanced

5. If $\begin{vmatrix} a_1+b_1x & a_1x+b_1 & c_1 \\ a_2+b_2x & a_2x+b_2 & c_2 \\ a_3+b_3x & a_3x+b_3 & c_3 \end{vmatrix}=0$, then possible conditions is/are -

(A) $x = 1 \ \forall \ a_i, b_i$, where $1 \le i, j \le 3$ (B) $x = -1 \ \forall \ a_i, b_i$, where $1 \le i, j \le 3$

(C) $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}=0$ (D) $x = \pm 2 \ \forall \ a_i, b_i$, where $1 \le i, j \le 3$

MDT158

6. $\begin{vmatrix} x^2 & (y+z)^2 & yz \\ y^2 & (x+z)^2 & zx \\ z^2 & (x+y)^2 & xy \end{vmatrix}$ is divisible by -

(A) $x^2 + y^2 + z^2$ (B) $x - y$ (C) $x - y - z$ (D) $x + y + z$

MDT159

7. If $\begin{vmatrix} \frac{a}{b}+\frac{a}{c} & \frac{b}{c}+\frac{b}{a} & 1 \\ \frac{b}{a}+\frac{b}{c} & \frac{c}{a}+\frac{c}{b} & 1 \\ \frac{c}{a}+\frac{c}{b} & \frac{a}{b}+\frac{a}{c} & 1 \end{vmatrix}=0$, where $a, b, c \in R^+$, then which of the following is necessarily true -

(A) $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=0$ (B) $a^2+b^2+c^2=ab+bc+ac$

(C) $a = b = c$ (D) $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=1$

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MDT160

8. If a, b, c are in A.P. and α, β, γ are positive real numbers in G.P., then the equation $\begin{vmatrix} x+a & x^2+\log\alpha & k \\ x+b & x^2+\log\beta & k \\ x+c & x^2+\log\gamma & k \end{vmatrix}=0$:-

(A) is an identity (B) has a root $x = 1$

(C) has a root $x = 0$ (D) has real & identical roots

MDT161

9. If α, β, γ satisfy the equation $\begin{vmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{vmatrix}=0$, then

(A) $\alpha + \beta + \gamma = 0$ (B) $\alpha^2 + \beta^2 + \gamma^2 = 6$ (C) $\alpha^3 + \beta^3 + \gamma^3 = -6$ (D) $\alpha^4 + \beta^4 + \gamma^4 = 18$

MDT162

10. If $\ell < m < n$, then $\begin{vmatrix} 1 & \ell & \ell^4 \\ 1 & m & m^4 \\ 1 & n & n^4 \end{vmatrix}$ will always be greater than -

(A) 3 (B) 6 (C) –1 (D) 0

MDT163

For More Join: @IITJEE_Advanced

11. Let $D_k=\begin{vmatrix}1 & 2k & 2k-1\\ n & n^2+n+2 & n^2\\ n & n^2+n & n^2+n+2\end{vmatrix}$. If $\sum_{k=1}^{n} D_k = 96$, then n is divisible by

(A) 2 (B) 3 (C) 4 (D) 6

MDT164

12. Let $\begin{vmatrix}a & \sqrt{5} & \sqrt{7}\\ \sqrt{3} & b & \sqrt{7}\\ \sqrt{3} & \sqrt{5} & c\end{vmatrix}=0$, $\left(a\neq\sqrt{3}, b\neq\sqrt{5}, c\neq\sqrt{7}\right)$ and $\frac{a}{a-\sqrt{3}}+\frac{b}{b-\sqrt{5}}+\frac{c}{c-\sqrt{7}}=\lambda$, then

(A) $\lambda = 2$ (B) $\lambda = 3$

(C) for $a=2\sqrt{3}$, $\frac{\sqrt{5}}{b}+\frac{\sqrt{7}}{c}=2$ (D) for $c=2\sqrt{7}$, $\frac{\sqrt{3}}{a}+\frac{\sqrt{5}}{b}=2$

MDT165

13. Let p and $p + 2$ be prime numbers and let $\Delta=\begin{vmatrix}p! & (p+1)! & (p+2)!\\ (p+1)! & (p+2)! & (p+3)!\\ (p+2)! & (p+3)! & (p+4)!\end{vmatrix}$.

If the sum of the maximum values of α and β, such that p^{α} and $(p + 2)^{\beta}$ divide Δ, is n then n divides

(A) 2 (B) 4 (C) 24 (D) 36

MDT166

14. If the system of equations $x + y - 3 = 0$, $(1 + K)x + (2 + K)y - 8 = 0$ & $x - (1 + K)y + (2 + K) = 0$ is consistent then the value of K may be -

(A) 1 (B) $\frac{3}{5}$ (C) $-\frac{5}{3}$ (D) 2

MDT167

15. If system of equation $a_1x + b_1y = c_1$ & $a_2x + b_2y = c_2$ (where $a_1, b_1, c_1, a_2, b_2, c_2 \neq 0$) has infinite solutions, then-

(A) $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$

(B) $\frac{a_1+a_2}{a_1-a_2}=\frac{b_1+b_2}{b_1-b_2}=\frac{c_1+c_2}{c_1-c_2}$

(C) the quadratic equations $a_1x^2 + b_1x + c_1 = 0$ & $a_2x^2 + b_2x + c_2 = 0$ have no common root

(D) system of equation $a_1^2\, a_2x + b_1^2b_2y = c_1^2c_2$ & $a_1a_2^2x + b_1b_2^2y = c_1c_2^2$ will also have infinite number of solutions

MDT168

16. The system of linear equations $x + y + z = 6$, $x + 2y + 3z = 14$ and $2x + 5y + pz = q$ have -

(A) infinitely many solution when $p = 8$, $q = 36$

(B) unique solution when $p \neq 8$, $q \neq 36$

(C) no solution when $p = 8$, $q \neq 36$

(D) atleast one solution for $q = 36$, $p \in R$

MDT169

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17. System of equation $x + y + az = b$, $2x + 3y = 2a$ & $3x + 4y + a^2z = ab + 2$ has

(A) unique solution when $a \neq 0$, $b \in R$

(B) no solution when $a = 0$, $b = 1$

(C) infinite solution when $a = 0$, $b = 2$

(D) infinite solution when $a = 1$, $b \in R$

MDT170

18. System of linear equations in x, y, z have infinite solutions which

$2x + y + z = 1$

$x - 2y + z = 2$

$3x - y + 2z = 3$

(A) can be written as $(-3\lambda - 1, \lambda, 5\lambda + 3)\ \forall\ \lambda \in R$

(B) can be written as $(3\lambda - 1, -\lambda, -5\lambda + 3)\ \forall\ \lambda \in R$

(C) are such that every solution satisfy $x - 3y + 1 = 0$

(D) are such that none of them satisfy $5x + 3z = 1$

MDT171

19. For the system of linear equation

$2x - y + 3z = 5$

$3x + 2y - z = 7$

$4x + 5y + \alpha z = \beta$

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Which of the following is(are) CORRECT ?

(A) The system has infinitely many solutions for $\alpha = -5$ and $\beta = 9$

(B) The system has a unique solution for $\alpha \neq -5$ and $\beta = 8$

(C) The system has infinitely many solutions for $\alpha = -6$ and $\beta = 9$

(D) The system is inconsistent for $\alpha = -5$ and $\beta = 8$

MDT172

20. If the system of equation

$x + \left(\sqrt{2}\sin\alpha\right)y + \left(\sqrt{2}\cos\alpha\right)z = 0$

$x + (\cos\alpha)y + (\sin\alpha)z = 0$

$x + (\sin\alpha)y - (\cos\alpha)z = 0$

has a non-trivial solution, then $\alpha =$

(A) $\frac{5\pi}{12}$ (B) $\frac{5\pi}{24}$ (C) $\frac{29\pi}{24}$ (D) $\frac{13\pi}{24}$

MDT173

For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

Paragraph for Question 21 to 23

Let $x, y, z \in \mathbb{R}^+$ & $D = \begin{vmatrix} x & x^3 & x^4-1 \\ y & y^3 & y^4-1 \\ z & z^3 & z^4-1 \end{vmatrix}$

On the basis of above information, answer the following questions :

21. If $x \neq y \neq z$ & x, y, z are in GP and $D = 0$, then y is equal to -

(A) 1 (B) 2 (C) 4 (D) none of these

MDT174

22. If x, y, z are the roots of $t^3 - 21t^2 + bt - 343 = 0$, $b \in R$, then D is equal to-

(A) 1 (B) 0

(C) dependent on x, y, z (D) data inadequate

MDT175

23. If $x \neq y \neq z$ & x, y, z are in A.P. and $D = 0$, then $2xy^2z + x^2z^2$ is equal to-

(A) 1 (B) 2 (C) 3 (D) none of these

MDT176

Paragraph for Question 24 to 26

Let $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ \alpha & \beta & \gamma \end{vmatrix} = t$, where t is an even prime number & α, β, γ are the integral roots of the equation

$x^3 - 14x^2 + Px - 36 = 0$

On the basis of above information answer the following :

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24. The value of P is -

(A) a rational number (B) a prime number

(C) an odd natural number (D) an even natural number

MDT177

25. The value of $\alpha(\beta^2+\gamma^2)+\beta(\gamma^2+\alpha^2)+\gamma(\alpha^2+\beta^2)$ is divisible by -

(A) 17 (B) 34 (C) 51 (D) 68

MDT178

26. Which of following statement is/are false -

(A) t is divisible by $(\alpha - \beta)$ (B) t is divisible by $(\beta - \gamma)$

(C) t is divisible by $(\gamma - \alpha)$ (D) $(\gamma - \alpha)$ is divisible by t

MDT179

Paragraph for Question 27 & 28

Let $f(x) = \begin{vmatrix} 1 & x-1 & x \\ 2(x-1) & (x-1)(x-2) & x(x-1) \\ 3(x-1)(x-2) & (x-1)(x-2)(x-3) & x(x-1)(x-2) \end{vmatrix}$ & $D_r = \begin{vmatrix} 1 & 10 & 2r \\ 70 & 17 & 3r+1 \\ 1 & 1 & 1 \end{vmatrix}$.

On the basis of above information, answer the following questions :

27. The value of $f(50) - D_5$ is -

MDT180

28. $\sum_{r=1}^{10} D_r$ is

MDT181

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTION

29. Consider a system of linear equations $a_i x + b_i y + c_i z = d_i$ (where $a_i, b_i, c_i \neq 0$ and $i = 1,2,3$) & (α,β,γ) is its unique solution, then match list-I with list-II

	List-I		**List-II**
(I)	If $a_i = k$, $d_i = k^2$, $(k \neq 0)$ and $\alpha + \beta + \gamma = 2$, then k is	(P)	1
(II)	If $a_i = d_i = k \neq 0$, then $\alpha + \beta + \gamma$ is	(Q)	2
(III)	If $a_i = k > 0$, $d_i = k + 1$, then $\alpha + \beta + \gamma$ can be	(R)	0
(IV)	If $a_i = k < 0$, $d_i = k + 1$, then $\alpha + \beta + \gamma$ can be	(S)	3
		(T)	–1

(A) I $\rightarrow P,Q$; II $\rightarrow R$; III $\rightarrow Q,S$; IV $\rightarrow T$ (B) I $\rightarrow P$; II $\rightarrow Q$; III $\rightarrow R,S$; IV $\rightarrow T$

(C) I $\rightarrow Q$; II $\rightarrow P,R$; III $\rightarrow S$; IV $\rightarrow T$ (D) I $\rightarrow Q$; II $\rightarrow P$; III $\rightarrow Q,S$; IV $\rightarrow R,T$

MDT182

30. Match the following for the system of linear equations

$\lambda x + y + z = 1, x + \lambda y + z = \lambda, x + y + \lambda z = \lambda^2$

	Column-I		**Column-II**
(A)	$\lambda = 1$	(P)	unique solution
(B)	$\lambda \neq 1$	(Q)	infinite solutions
(C)	$\lambda \neq 1, \lambda \neq -2$	(R)	no solution
(D)	$\lambda = -2$	(S)	finite many solutions

MDT183

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE (S)

1. Let a, b, c are the solutions of the cubic $x^3 - 5x^2 + 3x - 1 = 0$, then find the value of the determinant $\begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix}$.

MDT184

2. If $\Delta(x) = \begin{vmatrix} 0 & 2x-2 & 2x+8 \\ x-1 & 4 & x^2+7 \\ 0 & 0 & x+4 \end{vmatrix}$ and $f(x) = \sum_{j=1}^{3}\sum_{i=1}^{3} a_{ij}c_{ij}$, where a_{ij} is the element of ith and j^{th} column in $\Delta(x)$ and c_{ij} is the cofactor a_{ij} $\forall$ i and j, then find the greatest value of $f(x)$, where $x \in [-3, 18]$

MDT185

3. For a determinant Δ of order 3, the element a_{ij} is defined as $a_{ij} = \tan^{-1}(\tan(i - j))$ $\forall$ i, j, then the value of Δ is equal to (where 'i' represents row and 'j' represents column)

MDT186

4. Let $a, b, c, \ell, m, n \in R$ such that $a\ell + bm + cn = 0$, $b\ell + cm + an = 0$, $c\ell + am + bn = 0$. If a, b & c are distinct & $f(x) = ax^3 + bx^2 + cx + 5$, then the value of $f(1)$ is

MDT187

Rankers Academy JEE

5. If $\begin{vmatrix} \sin^2 2x & \cos 2x & 4\sin^2\frac{x}{2} \\ \tan^2\frac{x}{2} & \cos^2 x & -\sin^2 x \\ -2\cos 4x & \tan^2\frac{x}{2} & \sin 4x \end{vmatrix} = a_0 + a_1(\cos x) + a_2(\cos^2 x) + \ldots\ldots + a_n(\cos^n x)$, then a_0 is –

MDT188

6. If x, y, z are distinct digits $(0 \le x, y, z \le 9)$ & the minimum possible value of $\begin{vmatrix} z & 9y & x \\ z & y & 9x \\ 9z & y & x \end{vmatrix}$ is λ then $\frac{\lambda}{83700}$ is

(where $9x, 9y$ & $9z$ are two digits number)

MDT189

7. $\begin{vmatrix} a & b-c & c+b \\ a+c & b & c-a \\ a-b & b+a & c \end{vmatrix} = \begin{vmatrix} b-c & b & b+a \\ a & -a-c & b-a \\ c+b & c-a & c \end{vmatrix}$ and $a > 0, b > 0, c > 0$, then the minimum value of $\frac{c^2}{ab}$ is

MDT190

8. Find the sum of all positive integral values of a for which every solution to the system of equation $x + ay = 3$ and $ax + 4y = 6$ satisfy the inequalities $x > 1, y > 0$.

MDT191

For More Join: @IITJEE_Advanced

9. The number of triplets (α, β, γ) satisfying the following constraints

$2\alpha - \beta + 3\gamma = 4$

$\alpha + \beta - 3\gamma = -1$

$5\alpha - \beta + 3\gamma = 7$

$\alpha\beta\gamma \leq 0$

& $\alpha, \beta, \gamma \in I$

MDT192

10. If the equations $a(y + z) = x$, $b(z + x) = y$, $c(x + y) = z$ (where $a, b, c \neq -1$) have nontrivial solutions, then find the value of $\frac{1}{1+a}+\frac{1}{1+b}+\frac{1}{1+c}$.

MDT193

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. If $\alpha, \beta \neq 0$, and $f(n) = \alpha^n + \beta^n$ and $\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = K(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$, then K is equal to : **[JEE (Main) 2014]**

(1) $\alpha\beta$ (2) $\frac{1}{\alpha\beta}$ (3) 1 (4) –1

MDT194

2. The set of all values of λ for which the system of linear equations :
$2x_1 - 2x_2 + x_3 = \lambda x_1$, $2x_1 - 3x_2 + 2x_3 = \lambda x_2$, $-x_1 + 2x_2 = \lambda x_3$
has a non-trivial solution **[JEE (Main) 2015]**

(1) contains two elements (2) contains more than two elements
(3) is an empty set (4) is a singleton

MDT195

3. The system of linear equations $x + \lambda y - z = 0$, $\lambda x - y - z = 0$, $x + y - \lambda z = 0$ has a non-trivial solution for : **[JEE (Main) 2016]**

(1) exactly three values of λ. (2) infinitely many values of λ.
(3) exactly one value of λ. (4) exactly two values of λ.

MDT196

4. If S is the set of distinct values of 'b' for which the following system of linear equations

$$x + y + z = 1$$
$$x + ay + z = 1$$
$$ax + by + z = 0$$

has no solution, then S is : **[JEE (Main) 2017]**

(1) a singleton
(2) an empty set
(3) an infinite set
(4) a finite set containing two or more elements

MDT197

5. If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A + Bx)(x - A)^2$, then the ordered pair (A, B) is equal to :

[JEE (Main) 2018]

(1) (–4, 3) (2) (–4, 5) (3) (4, 5) (4) (–4, –5)

MDT198

6. If the system of linear equations

$$x + ky + 3z = 0$$
$$3x + ky - 2z = 0$$
$$2x + 4y - 3z = 0$$

has a non-zero solution (x, y, z), then $\frac{xz}{y^2}$ is equal to : **[JEE (Main) 2018]**

(1) 10 (2) – 30 (3) 30 (4) –10

MDT199

Rankers Academy JEE

For More Join: @IITJEE_Advanced

7. If the system of linear equations

$x - 4y + 7z = g$

$3y - 5z = h$

$-2x + 5y - 9z = k$

is consistent, then : **[JEE(Main) 2019]**

(1) $g + h + k = 0$ (2) $2g + h + k = 0$ (3) $g + h + 2k = 0$ (4) $g + 2h + k = 0$

MDT200

8. Let $d \in R$, and $A=\begin{bmatrix} -2 & 4+d & (\sin\theta)-2 \\ 1 & (\sin\theta)+2 & d \\ 5 & (2\sin\theta)-d & (-\sin\theta)+2+2d \end{bmatrix}$, $\theta \in [0, 2\pi]$. If the minimum value of $\det(A)$ is 8, then a value of d is : **[JEE (Main) 2019]**

(1) -7 (2) $2\left(\sqrt{2}+2\right)$ (3) -5 (4) $2\left(\sqrt{2}+1\right)$

MDT201

9. Let $a_1, a_2, a_3, ..., a_{10}$ be in G.P. with $a_i > 0$ for $i = 1,2,...., 10$ and S be the set of pairs (r, k), $r, k \in N$ (the set of natural numbers) for which $\begin{vmatrix} \log_e a_1^r a_2^k & \log_e a_2^r a_3^k & \log_e a_3^r a_4^k \\ \log_e a_4^r a_5^k & \log_e a_5^r a_6^k & \log_e a_6^r a_7^k \\ \log_e a_7^r a_8^k & \log_e a_8^r a_9^k & \log_e a_9^r a_{10}^k \end{vmatrix} = 0$. Then the number of elements in S, is : **[JEE (Main) 2019]**

(1) Infinitely many (2) 4 (3) 10 (4) 2

MDT202

Rankers Academy JEE

10. If the system of linear equations

$2x + 2ay + az = 0$

$2x + 3by + bz = 0$

$2x + 4cy + cz = 0,$

where $a, b, c \in R$ are non-zero and distinct; has a non-zero solution, then : **[JEE (Main) 2020]**

(1) a, b, c are in A.P. (2) $a + b + c = 0$

(3) a, b, c are in G.P. (4) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

MDT203

11. For which of the following ordered pairs (μ, δ), the system of linear equations

$x + 2y + 3z = 1$

$3x + 4y + 5z = \mu$

$4x + 4y + 4z = \delta$

is inconsistent ? **[JEE (Main) 2020]**

(1) (1,0) (2) (4,6) (3) (3,4) (4) (4,3)

MDT204

12. Let $a - 2b + c = 1$. If $f(x)=\begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$, then : **[JEE (Main) 2020]**

(1) $f(-50) = 501$ (2) $f(-50) = -1$ (3) $f(50) = 1$ (4) $f(50) = -501$

MDT205

For More Join: @IITJEE_Advanced

13. Let α, β, γ be the real roots of the equation, $x^3 + ax^2 + bx + c = 0$, $(a, b, c \in R$ and $a, b \neq 0)$. If the system of equations (in, u, v, w) given by $\alpha u + \beta v + \gamma w = 0$, $\beta u + \gamma v + \alpha w = 0$; $\gamma u + \alpha v + \beta w = 0$ has non-trivial solution, then the value of $\frac{a^2}{b}$ is **[JEE (Main) 2021]**

(1) 5 (2) 3 (3) 1 (4) 0

MDT206

14. Let the system of linear equations

$4x + \lambda y + 2z = 0$

$2x - y + z = 0$

$\mu x + 2y + 3z = 0, \lambda, \mu \in R.$

has a non-trivial solution. Then which of the following is true ? **[JEE (Main) 2021]**

(1) $\mu = 6, \lambda \in R$ (2) $\lambda = 2, \mu \in R$ (3) $\mu = 3, \mu \in R$ (4) $\mu = -6, \lambda \in R$

MDT207

15. The number of real values λ, such that the system of linear equations

$2x - 3y + 5z = 9$

$x + 3y - z = -18$

$3x - y + (\lambda^2 - |\lambda|)z = 16$

has no solution, is :- **[JEE (Main) 2022]**

(1) 0 (2) 1 (3) 2 (4) 4

MDT208

16. If the system of equations

$x + y + z = 6$

$2x + 5y + \alpha z = \beta$

$x + 2y + 3z = 14$

has infinitely many solutions, then $\alpha + \beta$ is equal to : **[JEE (Main) 2022]**

(1) 8 (2) 36 (3) 44 (4) 48

MDT035

Rankers Academy JEE

17. Let $f(x) = \begin{vmatrix} 1+\sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1+\cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1+\sin 2x \end{vmatrix}$, $x \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$. If α a β respectively are the maximum and the minimum values of f, then **[JEE (Main) 2023]**

(1) $\beta^2 - 2\sqrt{\alpha} = \frac{19}{4}$ (2) $\beta^2 + 2\sqrt{\alpha} = \frac{19}{4}$ (3) $\alpha^2 - \beta^2 = 4\sqrt{3}$ (4) $\alpha^2 + \beta^2 = \frac{9}{2}$

MDT108

18. Let S_1 and S_2 be respectively the sets of all $a \in R - \{0\}$ for which the system of linear equations

$ax + 2ay - 3az = 1$

$(2a+1)x + (2a+3)y + (a+1)z = 2$

$(3a+5)x + (a+5)y + (a+2)z = 3$

has unique solution and infinitely many solutions. Then **[JEE (Main) 2023]**

(1) $n(S_1) = 2$ and S_2 is an infinite set (2) S_1 is an infinite set an $n(S_2) = 2$

(3) $S_1 = \phi$ and $S_2 = \mathbb{R} - \{0\}$ (4) $S_1 = \mathbb{R} - \{0\}$ and $S_2 = \phi$

MDT111

For More Join: @IITJEE_Advanced

19. The values of α, for which $\begin{vmatrix} 1 & \frac{3}{2} & \alpha+\frac{3}{2} \\ 1 & \frac{1}{3} & \alpha+\frac{1}{3} \\ 2\alpha+3 & 3\alpha+1 & 0 \end{vmatrix}=0$, lie in the interval **[JEE (Main) 2024]**

(1) $(-2,1)$ (2) $(-3,0)$ (3) $\left(-\frac{3}{2},\frac{3}{2}\right)$ (4) $(0,3)$

MDT209

20. Consider the system of linear equation $x + y + z = 4\mu$, $x + 2y + 2\lambda z = 10\mu$, $x + 3y + 4\lambda^2 z = \mu^2 + 15$ where $\lambda, \mu \in R$. Which one of the following statements is NOT correct ? **[JEE (Main) 2024]**

(1) The system has unique solution if $\lambda \neq \frac{1}{2}$ and $\mu \neq 1,15$

(2) The system is inconsistent if $\lambda = \frac{1}{2}$ and $\mu \neq 1$

(3) The system has infinite number of solution if $\lambda = \frac{1}{2}$ and $\mu = 15$

(4) The system is consistent if $\lambda \neq \frac{1}{2}$

MDT210

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. The number of all possible values of θ, where $0 < \theta < \pi$, for which the system of equations

$(y + z)\cos 3\theta = (xyz)\sin 3\theta$

$x\sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$

$(xyz)\sin 3\theta = (y + 2z)\cos 3\theta + y\sin 3\theta$

have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is **[IIT JEE 2010]**

MDT211

2. Which of the following values of α satisfy the equation $\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha$?

[JEE (Advanced) 2015]

(A) – 4 (B) 9 (C) – 9 (D) 4

MDT054

Rankers Academy JEE

3. The total number of distinct $x \in R$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ is

[JEE (Advanced) 2016]

MDT055

4. Let $a, \lambda, \mu \in R$. Consider the system of linear equations

$ax + 2y = \lambda$

$3x - 2y = \mu$

Which of the following statement(s) is(are) correct ? **[JEE Advanced 2016]**

(A) if $a = -3$, then the system has infinitely many solutions for all values of λ and μ

(B) If $a \neq -3$, then the system has a a unique solution for all values of λ and μ

(C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$

(D) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$

MDT056

For More Join: @IITJEE_Advanced

5. Let p, q, r be nonzero real numbers that are, respectively, the 10th, 100th and 1000th terms of a harmonic progression. Consider the system of linear equations

$x + y + z = 1$
$10x + 100y + 1000z = 0$
$qrx + pry + pqz = 0.$

List-I		List-II	
(I)	If $\frac{q}{r} = 10$, then the system of linear equations has	(P)	$x=0,\ y=\frac{10}{9}, z=-\frac{1}{9}$ as a solution
(II)	If $\frac{p}{r} \neq 100$, then the system of linear equations has	(Q)	$x=\frac{10}{9},\ y=-\frac{1}{9}, z=0$ as a solution
(III)	If $\frac{p}{q} \neq 10$, then the system of linear equations has	(R)	infinitely many solutions
(IV)	If $\frac{p}{q} = 10$, then the system of linear equations has	(S)	no solution
		(T)	at least one solution

[JEE (Advanced) 2022]

The correct option is:
(A) (I) → (T); (II) → (R); (III) → (S); (IV) → (T)
(B) (I) → (Q); (II) → (S); (III) → (S); (IV) → (R)
(C) (I) → (Q); (II) → (R); (III) → (P); (IV) → (R)
(D) (I) → (T); (II) → (S); (III) → (P); (IV) → (T)

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MDT059

6. Let α, β and γ be real numbers. consider the following system of linear equations

$x + 2y + z = 7$
$x + \alpha z = 11$
$2x - 3y + \beta z = \gamma$

Match each entry in **List - I** to the correct entries in **List-II**

	List-I		List-II
(P)	If $\beta=\frac{1}{2}(7\alpha - 3)$ and $\gamma = 28$, then the system has	(1)	a unique solution
(Q)	If $\beta=\frac{1}{2}(7\alpha - 3)$ and $\gamma \neq 28$, then the system has	(2)	no solution
(R)	If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where $\alpha = 1$ and $\gamma \neq 28$, then the system has	(3)	infinitely many solutions
(S)	If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where $\alpha = 1$ and $\gamma = 28$, then the system has	(4)	$x = 11$, $y = -2$ and $z = 0$ as a solution
		(5)	$x = -15$, $y = 4$ and $z = 0$ as a solution

[JEE (Advanced) 2023]

For More Join: @IITJEE_Advanced

The correct option is :

(A) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (2) (R) $\rightarrow$ (1) (S) $\rightarrow$ (4)

(B) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (2) (R) $\rightarrow$ (5) (S) $\rightarrow$ (4)

(C) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (4) (S) $\rightarrow$ (5)

(D) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (1) (S) $\rightarrow$ (3)

MDT113

7. Let $S = \left\{ A = \begin{pmatrix} 0 & 1 & c \\ 1 & a & d \\ 1 & b & e \end{pmatrix} : a,b,c,d,e \in \{0,1\} \text{ and } |A| \in \{-1,1\} \right\}$, where $|A|$ denotes the determinant of A. Then the number of elements in S is _____. **[JEE (Advanced) 2024]**

MDT212

Rankers Academy JEE

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	C	A	B	D	B	C	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	D	D	B	C	B	A	C	D	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	A	D	C	B	D	A	A	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C,D	A,B,C,D	A,B,C,D	B,D	A,B,C	A,B,D	B,C	A,B,C	A,B,C,D	C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,D	A,C,D	B,C,D	A,C	A,B,D	A,B,C,D	B,C,D	A,B,D	A,B,D	B,C,D
Que.	21	22	23	24	25	26	27	28	29	
Ans.	A	B	C	A,C	A,B	A,B,C	9.00	575.00	D	
Que.	30									
Ans.	A → Q; B → P, R; C → P; D → R									

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EXERCISE (S)

1.	80	**2.**	0	**3.**	0	**4.**	5	**5.**	0
6.	9	**7.**	4	**8.**	4	**9.**	1	**10.**	2

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	1	1	2	1	2	3	1	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	2	1	3	3	1	4	2	2

EXERCISE - JEE (ADVANCED) PYQ

Que.	1	2	3	4	5	6	7
Ans.	3	B,C	2	B,C,D	B	A	16

For More Join: @IITJEE_Advanced

Important Notes

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For More Join: @IITJEE_Advanced